PHYS 1301 Lab 3: Motion, materials and particles

Name: _____ Section: _____ Date: _____

Lab partner: _____ Station: _____ Instructor initials: ____

Part A. A cart on a track

A motion sensor recorded the position of the cart every half second. Use the table to estimate the velocity and the acceleration of the cart. Each column gives the quantity named in its header; the derivatives were found by finite differences.

Table 1. Position, velocity and acceleration of the cart

Time t (s)	Position $x(t)$ (m)	Velocity $v = \frac{dx}{dt}$ (m/s)	Acceleration $a = \frac{d^2x}{dt^2}$ (m/s²)
0.0	0.000	0.00	0.40
0.5	0.050	0.20	0.40
1.0	0.200	0.40	0.40
1.5	0.450	0.60	0.40
2.0	0.800	0.80	0.40
2.5	1.250	1.00	0.40

The acceleration is constant, so the position grows as the square of the time: $x = \frac{1}{2}at^2$.

Part B. Elastic properties of materials

The wire you stretch in Part C is one of the materials below. Two of the columns share a heading: both moduli describe stiffness, one under tension and one under shear.

Table 2. Elastic moduli and density

Material	Modulus (GPa)		Density ρ (kg/m³)
	Young's E	Shear G	
Steel	200	80	7850
Aluminium	70	26	2700
Brass	100	37	8500
Copper	117	44	8960
Glass	65	26	2500

Part C. The particles in the wire

Each row of the next table is one property and each column one particle, so read it across: the first cell of a row names the property.

Table 3. Properties of the electron, proton and neutron

	Electron	Proton	Neutron
Charge (C)	-1.602×10^{-19}	1.602×10^{-19}	0
Mass (kg)	9.109×10^{-31}	1.673×10^{-27}	1.675×10^{-27}
Spin	1/2	1/2	1/2
Found in	Shells around the nucleus	Nucleus	Nucleus

Part D. Reference data

You will need the vapour pressure of water to correct the gas volume collected over water in Part E. The table runs over a page break; its header is repeated on the next page.

Table 4. Vapour pressure of water

Temperature (°C)	Pressure (kPa)	Pressure (mmHg)
0	0.611	4.58
4	0.813	6.10
8	1.073	8.05
12	1.403	10.52
16	1.818	13.63
20	2.339	17.54
24	2.985	22.39
28	3.782	28.37
32	4.759	35.70
36	5.947	44.61
40	7.384	55.38
44	9.112	68.35
48	11.17	83.79
52	13.62	102.2
56	16.51	123.8
60	19.93	149.5
64	23.93	179.5
68	28.59	214.4
72	34.00	255.0
76	40.24	301.8

Temperature (°C)	Pressure (kPa)	Pressure (mmHg)
80	47.41	355.6
84	55.62	417.2
88	64.98	487.4
92	75.65	567.4
96	87.73	658.0
100	101.3	760.0

Subtract the vapour pressure at the water temperature from the room pressure to find the pressure of the dry gas.